Module 6 Project Data Cleaning Assignment

Medi Cal access and provider capacity analysis

Business vision: A Medi-Cal access strategy manager at DHCS must decide which counties and language populations should be prioritized for additional language-access capacity.

GitHub submission information

Branch name	Branch URL	Cleaning workflow commit ID
fennie-project	https://github.com/jguo368/dso576/tree/fennie-project	794e5d6fcce8abe6a8a8fa9a0488ed71ec8ed5e3

Fourteen item cleaning checklist

#	Status	Evidence
1	Done	County-month-language eligible populations are a planning signal; provider capacity remains unresolved because no provider dataset is present.
2	Done	Grain and candidate keys are documented for all four datasets.
3	Done	Original CSVs are read-only; source hashes are recorded.
4	Done	Raw fields remain and parsed date/count/comparison fields are added; conversion failures are reported.
5	Done	Only outer-whitespace comparison fields are added; language categories are not combined.
6	Done	Exact duplicates and repeated candidate keys are reported separately.
7	Done	Missing counts and percentages are reported before and after; blanks stay missing.
8	Done	Negative values, failed parses, invalid counties, placeholders, and P records are flagged.
9	Done	One-to-one county-month key coverage is verified; long demographic tables are not directly merged.
10	Done	631,980 rows before and 631,980 rows after across the four files; no rows were dropped.
11	Done	Missing counts are never changed to zero; county rankings/maps filter Invalid County Code.
12	Done	Five difficult individual-record checks all passed.
13	Done	A restart/rerun produced identical non-manifest output hashes.
14	Unresolved	Provider capacity, F/P definitions, suppression policy, and merged-file provenance need documentation.

Cleaning decision log

Rule	Reason	Validation check
Preserve original CSVs; write only derived outputs.	Protects source evidence and makes reruns safe.	Input hashes are saved; script has no source-file write path.
Keep blank/suppressed counts as missing.	Annotations indicate suppression; zero would invent a count.	Check parsed blanks remain missing and annotations remain present.
Create stripped comparison fields only.	Outer whitespace should not split comparisons; raw categories remain auditable.	Compare raw/clean mappings; no language recoding.
Keep F and P rows and flag P.	Status definition is unresolved.	Row-count audit reports P rows; neither status is filtered.
Keep and flag Invalid County Code rows.	They cannot safely be assigned to a county.	Report retained rows; exclude only in downstream ranking/map filters.
Do not row-level merge demographic long tables.	Different grains would multiply rows.	Key-validation report documents aggregate-first requirement.

Five individual record checks

Case	Original	Expected	Actual	Match
Whitespace in Language	' Unknown'	Raw language retained; comparison field equals stripped raw value.	raw=' Unknown'; clean='Unknown'	True
Blank/suppressed Total Eligibles	Total Eligibles=''; annotation='Cell suppressed for small numbers'	Parsed count is missing, not zero; annotation is retained.	parsed=<NA>; annotation='Cell suppressed for small numbers'	True
Date of Data placeholder	' .'	Raw placeholder retained; parsed date missing and placeholder flag True.	parsed_missing=True; flag=True	True
Invalid County Code	Invalid County Code	Record retained and invalid-county flag True; not treated as a normal county.	retained=True; invalid_county_flag=True	True
Derived-file key validation	source keys=11603; derived keys=11603	One-to-one key validation passes with no unmatched keys or row multiplication.	matched=11603; source_only=0; derived_only=0; none: validate=one_to_one passed	True

Codex Suggestion Verification

Suggestion checked: Retain and flag Invalid County Code records instead of deleting them. The original derived county-month file contains three affected records. All three remain in the cleaned output and carry is_invalid_county_code=True, so cleaning preserves row count and audit visibility. Because Invalid County Code is not a California county, county-level rankings and maps must exclude only those flagged records. The derived-file key check matched all 11,603 non-statewide county-month keys with zero unmatched keys and no row multiplication.

Reproducibility and limits

Running python cleaning/clean_medi_cal.py loads only original CSV inputs and writes derived outputs. A second run produced identical non-manifest output hashes. Blank and suppressed counts remain missing; F and P records remain present with a P flag. Provider-capacity values and definitions are not invented because no provider dataset is in the inspected files.